THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 13 & 14: **Two Variables, One Line, Math of Space: Surface Area and Volume** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** a straight line	plotted solutions lie on a line
2	**(b)** c = 0	x = 0, y = 0 satisfies ax + by = 0
3	**(d)** A is false but R is true	A is false (slope is undefined); R is true
4	**(a)** (4, -1)	substitute the pair in both equations
5	**(d)** 8	solve the pair first
6	**(b)** 256 cm^2	$2(lw + wh + lh) = 2(60 + 20 + 48) = 256$
7	**(d)** 360 cm^3	$lwh = 15 \times 6 \times 4 = 360$
8	**(d)** 440 cm^2	$2\pi rh = 440$
9	**(a)** 1848 cm^3	$\pi r^2 h = 1848$
10	**(a)** 1232 cm^3	$\frac{1}{3}\pi r^2 h = 1232$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	For 6 km: $x + 6y = 110$ (1). For 10 km: $x + 10y = 170$ (1).	1+1
12	Ratios: $\frac{2}{4} = \frac{3}{6} = \frac{9}{18} = \frac{1}{2}$ (1). The lines coincide, so there are infinitely many solutions (1).	1+1
13	$V_{cone} : V_{cyl} = \frac{1}{3}\pi r^2 h : \pi r^2 h$ (1) $= 1 : 3$ (1).	1+1
14	TSA $= 2\pi rh + 2\pi r^2 = 2\pi r(h + r)$ (1). $2\pi rh$ is the curved surface; $2\pi r^2$ is the two circular ends (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Let x be the number of ₹5 coins and y of ₹10 coins: $x + y = 30$ (1). $5x + 10y = 220$, i.e. $x + 2y = 44$ (1). Subtracting, $y = 14$ and $x = 16$; check $80 + 140 = 220$ (1).	1+1+1
16	Area $= 4\pi r^2 = 4 \times \frac{22}{7} \times 49$ (1) $= 616$ cm^2 (1). Cost $= 616 \times 0.25 =$ ₹154 (1).	1+1+1
17	(i) $212 = \frac{9}{5}C + 32$, so $\frac{9}{5}C = 180$ and $C = 100$ °C (1). (ii) Put $F = C$: $C = \frac{9}{5}C + 32$, so $-\frac{4}{5}C = 32$ (1). $C = -40$; the readings are equal at -40 (1).	1+1+1
	OR: Adding: $20x + 20y = 140$, so $x + y = 7$ (1). Subtracting: $2x - 2y = 6$, so $x - y = 3$ (1). Then $x = 5$ and $y = 2$; check $55 + 18 = 73$ (1).	1+1+1
18	(i) $3^2 : 5^2 = 9 : 25$ (1.5). (ii) $3^3 : 5^3 = 27 : 125$ (1.5).	1½+1½
	OR: $V = \frac{2}{3}\pi r^3 = \frac{2}{3} \times \frac{22}{7} \times 10.5^3$ (1) $= 2425.5$ cm^3 (1) $= 2.4255$ litres (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Slope $= 4$: the level rises 4 cm every minute (1). (ii) $y = 4(6) = 24$ cm (1). (iii) $4x = 50$, $x = 12.5$ minutes (1). (iv) $4(8) = 32 \neq 30$, so $(8, 30)$ is not on the line (1). (v) 10 is the y-intercept: the water level is already 10 cm at the start (1).	1+1+1+1+1
	OR: (i) Slope $= \frac{6 - (-2)}{4 - 0} = 2$ (1). (ii) $y = 2x - 2$ (1). (iii) Put $y = 0$: $x = 1$, so $(1, 0)$ (1). (iv) $x + 2x - 2 = 4$, so $3x = 6$, $x = 2$ (1). $y = 2$, so they meet at $(2, 2)$; check $2 + 2 = 4$ (1).	1+1+1+1+1
20	(i) Cube $= 8^3 = 512$ cm^3; cuboid $= 16 \times 8 \times 4 = 512$ cm^3 (1). (ii) Cube TSA $= 6 \times 64 = 384$ cm^2; cuboid TSA $= 2(128 + 32 + 64) = 448$ cm^2 (1½). (iii) The cube needs less, by 64 cm^2 (1). (iv) $64 \times 0.5 \times 100 =$ ₹3200 (1). (v) Cooling or heat transfer (radiators), or absorption in biological cells (½).	1+1½+1+1+½
	OR: Cone $= \frac{1}{3}\pi r^2(2r) = \frac{2}{3}\pi r^3$ (1). Sphere $= \frac{4}{3}\pi r^3$ (1). Cylinder $= \pi r^2(2r) = 2\pi r^3$ (1). Dividing by $\frac{2}{3}\pi r^3$: $1 : 2 : 3$ (1). For $r = 3$: $18\pi : 36\pi : 54\pi = 1 : 2 : 3$ (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 13 & 14: **Two Variables, One Line, Math of Space: Surface Area and Volume** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 3/2	rise ÷ run
2	**(a)** -3	put x = 0
3	**(c)** infinitely many solutions	compare a1/a2, b1/b2, c1/c2
4	**(c)** 1	k/3 = 2/6 = 4/12 gives k = 1
5	**(a)** ₹300	2x + 3y = 1050, 3x + 2y = 1200 give x = 300, y = 150
6	**(b)** 616 cm^2	$4\pi r^2 = 616$
7	**(b)** 308 cm^2	$2\pi r^2$
8	**(a)** 144 cm^3	$\frac{1}{3} \times 48 \times 9 = 144$
9	**(b)** $2r$	$2\pi r \times 2r = 4\pi r^2$
10	**(c)** A is true but R is false	the cylinder has height $2r$; A is false

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\frac{3}{6} = \frac{2}{4} = \frac{1}{2}$ but $\frac{5}{11} \neq \frac{1}{2}$ (1). The lines are parallel, so there is no solution (1).	1+1
12	Slope $= \frac{3}{12} = \frac{1}{4}$ (1). For every 4 m moved horizontally the ramp rises 1 m (1).	1+1
13	Surface areas are in the ratio $r_1^2 : r_2^2$ (1) $= 1 : 4$ (1).	1+1
14	Surface area becomes 4 times (1); volume becomes 8 times (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Multiply by 3: $3y = 2x - 12$, so $2x - 3y - 12 = 0$ (1). At $(6, 0)$: $12 - 0 - 12 = 0$ (1). At $(3, -2)$: $6 + 6 - 12 = 0$ (1).	1+1+1
16	$2\pi r = 66 \Rightarrow r = 10.5$ m (1). CSA $= 2\pi r^2 = 2 \times \frac{22}{7} \times 110.25 = 693$ m^2 (1). Cost $= 693 \times 40 =$ ₹27,720 (1).	1+1+1
17	Slope $= \frac{1}{12} = \frac{30}{L}$ (1). $L = 30 \times 12 = 360$ cm (1). That is 3.6 m (1).	1+1+1
	OR: First equation: $2a - 3 = 1$, so $a = 2$ (1). Second: $8 - b = 5$, so $b = 3$ (1). Verify: $2(2) + 3(-1) = 1$ and $4(2) + 3(-1) = 5$ (1).	1+1+1
18	New edge $= 1.5a$ (1). (i) TSA becomes $2.25\times$ the original: increase 125% (1). (ii) Volume becomes $3.375\times$ the original: increase 237.5% (1).	1+1+1
	OR: Extra volume $= \pi(3)^2(4) = 36\pi$ cm^3 (1). Volume of one marble $= \frac{4}{3}\pi$ cm^3 (1). Number $= 36\pi \div \frac{4}{3}\pi = 27$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $2x + 3y = 80$, $3x + 2y = 70$ (1). (ii) Adding: $5x + 5y = 150$; subtracting: $x - y = -10$, so $x + y = 30$ and $x - y = -10$ give $x = 10$, $y = 20$ (2). (iii) ₹30 (1). (iv) Exactly one pair of prices fits both bills; the lines intersect at one point (1).	1+2+1+1
	OR: Let boat speed be x and stream speed y (km/h). Downstream speed $x + y = \frac{20}{2} = 10$ (1). Upstream speed $x - y = \frac{12}{2} = 6$ (1). Adding: $2x = 16$, $x = 8$ (1). Then $y = 2$ (1). Boat 8 km/h, stream 2 km/h; check $8 + 2 = 10$, $8 - 2 = 6$ (1).	1+1+1+1+1
20	(i) $V = \pi \times 9 \times 8 = 72\pi$ cm^3 (1). (ii) $\frac{4}{3}\pi(1.5)^3 = 4.5\pi$ cm^3 (1½). (iii) $72\pi \div 4.5\pi = 16$ (1). (iv) One sphere: $4\pi(2.25) = 9\pi$ cm^2 (1); total $= 16 \times 9\pi = 144\pi$ cm^2 (½).	1+1½+1+1½
	OR: New radius $= 1.2r$ (1). (i) CSA $= 2\pi rh$ becomes $1.2\times$: increase 20% (1). (ii) Volume $= \pi r^2 h$ becomes $1.44\times$: increase 44% (1). (iii) With $h = r$: old TSA $= 2\pi r(2r) = 4\pi r^2$; new TSA $= 2\pi(1.2r)(r + 1.2r) = 5.28\pi r^2$, an increase of 32% (2).	1+1+1+2

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ANSWER KEY & MARKING SCHEME SET C

Chapters 13 & 14: **Two Variables, One Line, Math of Space: Surface Area and Volume** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** parallel to the y-axis	x stays 5 while y varies
2	**(b)** rises as we move from left to right	rise and run have the same sign
3	**(c)** A is true but R is false	A is true (coincident lines); R is false because parallel lines give none
4	**(b)** (2, 5)	substitute the pair in both equations
5	**(d)** 3	rise ÷ run
6	**(d)** 190 cm^2	$2(lw + wh + lh) = 2(63 + 14 + 18) = 190$
7	**(a)** 54 cm^2	side $= 3$ cm, TSA $= 6a^2 = 54$
8	**(c)** 264 cm^2	$2\pi rh = 264$
9	**(d)** 308 cm^3	$\pi r^2 h = 308$
10	**(a)** 264 cm^3	$\frac{1}{3}\pi r^2 h = 264$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Multiplying $x + 2y = 4$ by 3 gives $3x + 6y = 12$ (1). So the two equations are multiples of each other and every solution of one is a solution of the other (1).	1+1
12	Substitute $x = 1$, $y = k$: $3 + 2k = 7$ (1). So $k = 2$ (1).	1+1
13	A cross-section is the shape obtained by taking a very thin slice of the solid (1). Parallel to the base the slice is a circle (1).	1+1
14	$V = \frac{1}{3} \times$ area of base $\times$ height $= \frac{1}{3} \times 25 \times 12$ (1) $= 100$ cm^3 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Slope of $l_1 = \dfrac{8-2}{3-0} = 2$ (1). Slope of $l_2 = \dfrac{0-(-4)}{2-0} = 2$ (1). Equal slopes with different y-intercepts (2 and -4): parallel lines, so no solution (1).	1+1+1
16	$lw \cdot wh \cdot lh = (lwh)^2 = 12 \times 20 \times 15 = 3600$ (2). So $V = lwh = 60$ cm^3 (1).	2+1
17	Suppose they meet at a point: then $mx + d_1 = mx + d_2$ (1). This gives $d_1 = d_2$ (1), which contradicts $d_1 \neq d_2$; so they have no common point and are parallel (1).	1+1+1
	OR: $2x + 3y = 1050$ and $3x + 2y = 1200$ (1). Adding: $5x + 5y = 2250$, so $x + y = 450$; subtracting: $x - y = 150$ (1). So $x = 300$ and $y = 150$ (1).	1+1+1
18	Area $= \pi r(2h + r)$ (1) $= \frac{22}{7} \times 3.5 \times (20 + 3.5)$ (1) $= 11 \times 23.5 = 258.5$ cm^2 (1).	1+1+1
	OR: Cylinder $= \frac{22}{7} \times 49 \times 12 = 1848$ cm^3 (1). Cone $= \frac{1}{3} \times 1848 = 616$ cm^3 (1). Difference $= 1232$ cm^3 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$: intersecting lines, unique solution (1). (ii) $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$: coincident lines, infinitely many solutions (1). (iii) $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$: parallel lines, no solution (1). For the pair: $\frac{4}{6} = \frac{-2}{-3} = \frac{2}{3}$ and $\frac{-6}{-10} = \frac{3}{5}$ (1). Since $\frac{2}{3} \neq \frac{3}{5}$ the lines are parallel and there is no solution (1).	1+1+1+1+1
	OR: Let (x_1, y_1) and (x_2, y_2) lie on the line: $y_1 = mx_1 + d$ and $y_2 = mx_2 + d$ (1). Slope $= \dfrac{y_2 - y_1}{x_2 - x_1} = \dfrac{m(x_2 - x_1)}{x_2 - x_1} = m$ (2). Rewrite $-3y = -6x - 9$, so $y = 2x + 3$ (1). Slope 2, y-intercept 3 (1).	1+2+1+1
20	(i) $V = 20 \times 10 \times 1.8 = 360$ m^3 = 3,60,000 litres (1). (ii) Floor $= 200$ m^2; walls $= 2(20 + 10) \times 1.8 = 108$ m^2; total $= 308$ m^2 (2). (iii) $308 \times 250 =$ ₹77,000 (1). (iv) $360000 \div 3000 = 120$ min $= 2$ h (1).	1+2+1+1
	OR: (i) $V = \frac{4}{3} \times \frac{22}{7} \times 3.5^3 = 179\frac{2}{3} \approx 179.67$ cm^3 (1). (ii) Rise $= \frac{179.67}{\pi \times 49} = \frac{179.67}{154} = \frac{7}{6} \approx 1.17$ cm (2). (iii) $3 \times \frac{7}{6} = 3.5$ cm (1). (iv) Displaced volume equals base area $\times$ rise; a larger base gives a smaller rise (1).	1+2+1+1

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ANSWER KEY & MARKING SCHEME SET D

Chapters 13 & 14: **Two Variables, One Line, Math of Space: Surface Area and Volume** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** -2	write as y = mx + d, so m = −a/b
2	**(a)** (2, 3)	substitute each point in the equation
3	**(c)** no solution	compare a1/a2, b1/b2, c1/c2
4	**(c)** 1	k/3 = 2/6 gives k = 1 and then 5/12 ≠ 1/3
5	**(d)** 74	x + y = 11, x − y = 3 give x = 7, y = 4
6	**(d)** 154 cm^2	$4\pi r^2 = 154$
7	**(b)** 1848 cm^2	$3\pi r^2$ (curved + base)
8	**(b)** one-third of	salt-filling experiment: three cones fill one cylinder
9	**(a)** Both A and R are true and R is the correct explanation of A	e.g. $8 \times 8 \times 8$ and $16 \times 8 \times 4$ both have volume 512 but TSA 384 and 448
10	**(b)** Both A and R are true but R is not the correct explanation of A	both true; the cuboid formula does not explain the pyramid formula

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Adding: $2x = 16$, so $x = 8$ (1). Then $y = 12 - 8 = 4$ (1).	1+1
12	Put $y = 0$: $x = 4$, so $(4, 0)$ on the x-axis (1). Put $x = 0$: $y = 3$, so $(0, 3)$ on the y-axis (1).	1+1
13	Volume $= 2 \times 1 \times 1.5 = 3$ m^3 (1) $= 3000$ litres (1).	1+1
14	Rotate the triangle through $360°$ about one of its two perpendicular sides (1); this side becomes the height, the other becomes the radius and the hypotenuse the slant height (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Let the numbers be x and y with $x = 3y + 4$ and $x + y = 40$ (1). So $4y + 4 = 40$, $y = 9$ (1). Then $x = 31$; check $31 + 9 = 40$ (1).	1+1+1
16	$V = 25 \times 10 \times 2 = 500$ m^3 (1) $= 5{,}00{,}000$ litres (1). Time $= \frac{500000}{2500} = 200$ min $= 3$ h 20 min (1).	1+1+1
17	Let it be $\frac{n}{d}$: $5n + 5 = 3d + 3$, so $5n - 3d = -2$ (1). $2n - 2 = d - 1$, so $d = 2n - 1$ (1). Then $5n - 6n + 3 = -2$, $n = 5$ and $d = 9$; the fraction is $\frac{5}{9}$ (1).	1+1+1
	OR: $y = 2x - 3$ (1). Put $y = 0$: $2x = 3$ (1). The line meets the x-axis at $\left(\frac{3}{2}, 0\right)$ (1).	1+1+1
18	CSA $= 2\pi r^2 = 2 \times \frac{22}{7} \times 196 = 1232$ cm^2 (1.5). TSA $= 3\pi r^2 = 1848$ cm^2 (1.5).	1½+1½
	OR: $6^2 + 8^2 = 10^2$, so the base is a right triangle (1) of area $\frac{1}{2} \times 6 \times 8 = 24$ cm^2 (1). $V = \frac{1}{3} \times 24 \times 9 = 72$ cm^3 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$l + w = 22$ (1). New sides equal: $l - 2 = w + 3$, so $l - w = 5$ (1). Adding: $2l = 27$, $l = 13.5$ cm (1). $w = 8.5$ cm (1). Area $= 13.5 \times 8.5 = 114.75$ cm^2 (1).	1+1+1+1+1
	OR: $8x + 5y = 190$ and $5x + 8y = 187$ (1). Adding: $13x + 13y = 377$, so $x + y = 29$ (1). Subtracting: $3x - 3y = 3$, so $x - y = 1$ (1). Hence $x = 15$ and $y = 14$: a mango costs ₹15 and a guava ₹14 (1). The coefficients of x and y are interchanged, so adding and subtracting gives two simple equations at once (1).	1+1+1+1+1
20	(i) Half the side is 8; $l = \sqrt{8^2 + 15^2} = 17$ cm (1). (ii) $4 \times \frac{1}{2} \times 16 \times 17 = 544$ cm^2 (1). (iii) $544 + 256 = 800$ cm^2 (1). (iv) $V = \frac{1}{3} \times 256 \times 15 = 1280$ cm^3 (1). (v) Surface area is just the sum of the areas of the faces (1).	1+1+1+1+1
	OR: (i) Each tiny region of area a_i with the centre forms a cone of height r and volume $\frac{1}{3}a_i r$ (1); adding all, $V = \frac{1}{3}r \times \sum a_i = \frac{1}{3}r \times 4\pi r^2$ (2) $= \frac{4}{3}\pi r^3$ (1). (ii) $4\pi r^2 = 1386 \Rightarrow r = 10.5$ cm, $V = \frac{1}{3} \times 1386 \times 10.5 = 4851$ cm^3 (1).	1+2+1+1

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